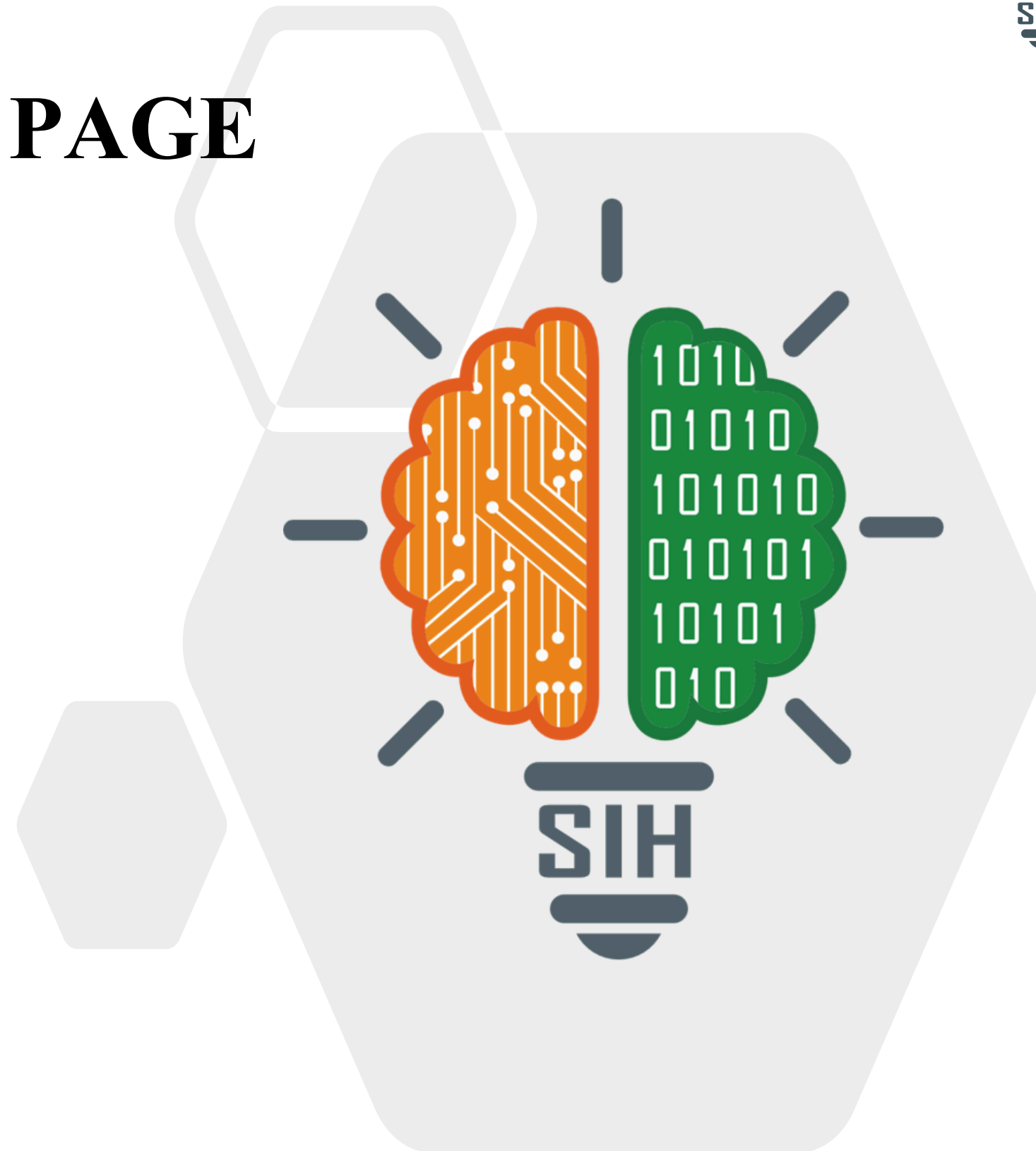


TITLE PAGE

- Problem Statement ID –
- Problem Statement Title-
- Theme-
- PS Category- Software/Hardware
- Team ID-
- Team Name (Registered on portal)



WHAT PROBLEM ARE WE SOLVING

-  **HUMAN BARRIER** - “Text intimidation & digital literacy” + “Language/Dialect Void”
Text-heavy systems and formal languages exclude beneficiaries; training may ignore their interests, traditional trades and mobility constraints.
-  **MARKET MISMATCH** - “Zero Hyper-Local Market Alignment”
Generic training ignores ODOP, District Industrial Profiles and local MSME demand, leading to poor placement.
-  **BUREAUCRATIC GAP** - “Lack of Perspective Plans” + “Post-Training Abandonment”
Insufficient ground data, financial support and placement tracking weaken the skilling-to-livelihood pipeline.

VOICE-FIRST PROFILING

Replaces intimidating text forms with natural, empathetic voice conversations.

Captures : Education & existing skills, Traditional family occupation, Physical mobility, Wage vs. self-employment preference

SMART PROFILING & MATCHING

Evaluates beneficiary voice inputs and maps them to 2,500+ official NSQF job roles.

Cross-references:

Beneficiary Aspirations + ODOP + Local MSMEs + District Job Vacancies

TRINITY-CHANNEL ACCESS

IVR — Toll-free feature-phone access

WhatsApp — Audio-in / audio-out for low-data users

Edge Kiosk — Offline PWA/Android app with local data queuing



KEY INNOVATIONS & USPs

Voice. AI. Inclusion.
Real Impact.

01

Dialect-Native Empathetic Voice Persona

Powered by MeitY Bhashini API + IndicLID

- Understands rural dialects (Bhojpuri, Bundeli, Chhattisgarhi, Malvi, etc.)
- Empathetic, human-like conversations mimicking a village elder (Gram Sahaayak)

02

Zero-Internet Edge AI Architecture

Runs quantized open-source AI models (Whisper-Tiny + Llama-3.2 1B on ONNX) locally

- Works on offline kiosks
- Designed for zero-connectivity aspirational blocks

03

Automated Government Perspective Plan Engine

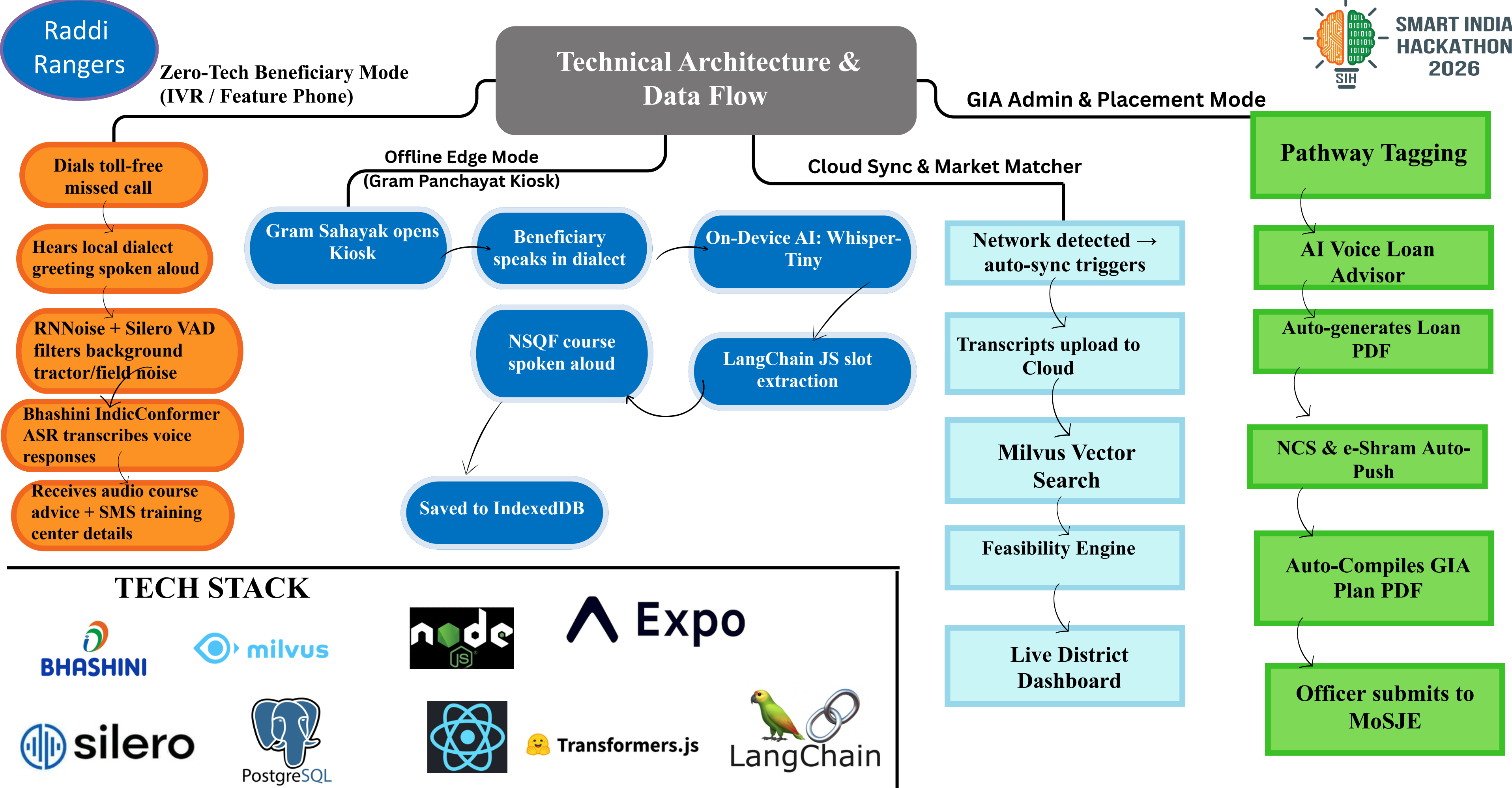
- Aggregates thousands of rural voice profiles
- Auto-creates ready-to-submit District Perspective Plan PDFs and budget forecasts for officers

04

End-to-End Livelihood Pipeline

Connects Beneficiary Voice to Real Opportunities





TECHNICAL FEASIBILITY

- Production-ready JS stack (React Native + Node.js)
- Proven Bhashini and AI APIs

OPERATIONAL FEASIBILITY

- **Trinity-Channel Reach:** Covers 100% of rural populations—smartphones via the **React Native App & WhatsApp**, and basic feature phones via **IVR toll-free calls**.

FINANCIAL FEASIBILITY

- Zero hardware cost for users
- **Scalable Infrastructure:** Low-cost cloud backend that integrates directly into existing **PM-AJAY GIA administrative budgets** without requiring custom hardware setup

POTENTIAL CHALLENGES AND RISKS

Poor cellular / internet network in remote aspirational blocks

Heavy rural background noise and local dialect speech variations.

Bank loan rejections for SC youth attempting self-employment.

STRATEGIES FOR OVERCOMING CHALLENGES

- Offline Edge PWA with IndexedDB queuing; auto-syncs when device catches internet. IVR PSIN fallback for feature phones

- Integrated RNNoise / Silvero VAD filtering + Bhashini ASR fallback to regional language models.

- AI Financial Consultant auto-generates 1-page bank-ready project reports for MUDRA / GIA 50,000 RS subsidies.

POTENTIAL IMPACT

- **100% Digital & Language Inclusivity:** Eliminates intimidation for low-literacy beneficiaries by replacing complex web forms with natural voice interaction in regional Indian languages and local dialects.
- **Aspiration-Driven Skilling:** Empowers rural SC youth by respecting their traditional family trades, physical mobility constraints, and personal career preferences rather than imposing arbitrary training quotas.
- **Zero Dropouts Due to Mismatch:** Beneficiaries enrolled in courses actually complete their training because the skilling aligns with their intrinsic interest and local market demand.

BENEFITS OF OUR SOLUTION

SOCIAL IMPACT

- **Dignity of Choice:** Gives marginalized beneficiaries a dignified, empathetic voice assistant (Gram Sahayak) that listens to their aspirations without administrative bias.
- **Women & Gender Inclusion:** Supports mobility-constrained women beneficiaries by recommending home-based micro-enterprise or hyper-local SHG livelihood pathways.

ECONOMIC IMPACT

- **3x Higher Post-Training Employment:** Cross-references beneficiary skills with One District One Product (ODOP) and local MSME clusters, guaranteeing actual post-training employment.
- **Financial Empowerment:** Bridges the financial gap by guiding self-employment candidates step-by-step through MUDRA / Stand Up India loans & GIA ₹50,000 subsidies.

ADMINISTRATIVE BENEFITS

- **Solves the Perspective Plan Bottleneck:** Auto-generates data-backed 3-to-5-year District GIA Perspective Plans for State SC Corporations in minutes instead of months.
- **Achieves Scheme Target:** Directly enables Ministry of Social Justice & Empowerment (MoSJE) to achieve its mandatory 70% wage/self-employment placement rate.
- **Inter-Agency Convergence:** Connects MoSJE, State SC Corporations, Training Providers, and Banks on a single transparent dashboard.

National Career Service (NCS) API Specifications: ncs.gov.in

e-Shram Portal & Unorganized Worker Data Standards: <https://eshram.gov.in/>

PM-AJAY Scheme Operational Guidelines: dosje.gov.in/schemes/PM-AJAY

National Skills Qualifications Framework (NSQF): ncvet.gov.in / ncvtm.gov.in

One District One Product (ODOP) & MSME Clusters: odop.dpiit.gov.in

MeitY Bhashini Open API & ULCA Architecture: bhashini.gov.in

Transformers.js & ONNX WebAssembly Runtime: huggingface.co/docs/transformers.js

AI4Bharat IndicConformer & Indic-TTS Research: ai4bharat.iitm.ac.in